Tutorial 3 Solutions - part 1

Exercise 1 Solutions

- **a. Degree = 5.**
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- **b. Possible Domains:**
  - `vehicle_id`: Strings composed of a fixed 3-letter classification prefix (`TRK` or `VAN`) followed by a hyphen and a 3-digit numerical identifier code.
  - `primary_driver`: Text strings representing the last name of the employee, followed by a comma, a space, and the initial letter of their first name completed with a period (e.g., `"Anderson, T."`).
  - `capacity_tons`: The set of strict positive integers representing structural cargo load capacity boundaries (e.g., values (>0)).

Exercise 2 Solutions

- **a. Vehicle Table Correction:** Isolate driver information into its own entity to eliminate update and insertion anomalies.
  - `Vehicle (vehicle_id, model_name, capacity_tons, driver_id, last_maintenance)`
  - `Driver (driver_id, Last_Name, First_Name_Initial)`
- **b. Medical_Clinics Table Correction:** Atomic splitting of combined geographical data elements and structural person components.
  - `Clinics (Clinic_ID, Clinic_Name, City, State, Physician_ID)`
  - `Physicians (Physician_ID, First_Name, Last_Name)`
- **c. Corporate_Assets Table Correction:** Cleanse currency attributes to hold homogeneous unit calculations.
  - `Assets (Asset_ID, Asset_Name, Description, Unit_Value_USD)`